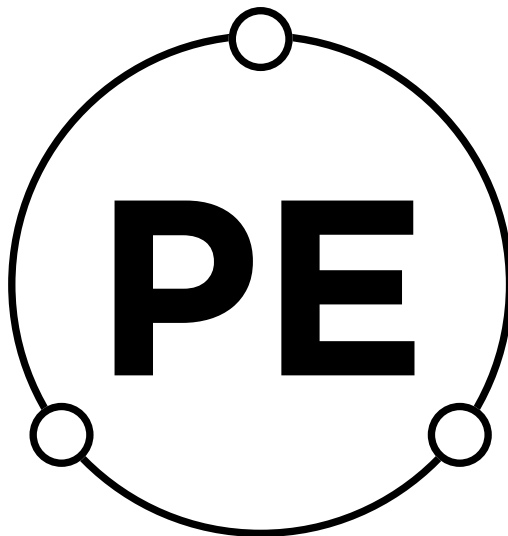


POUNAMU ELECTRONICS
RE_SWC
USER GUIDE

Rev 4

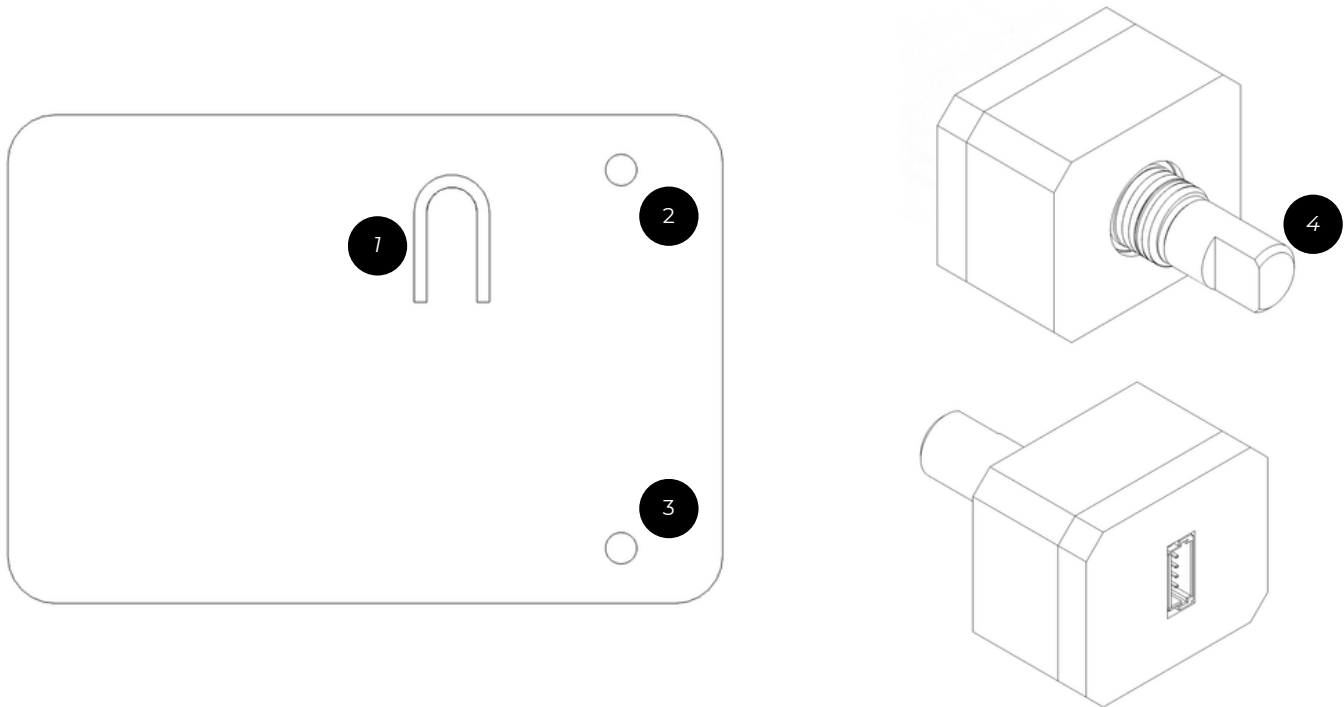


**POUNAMU
ELECTRONICS**

Contents

3	<u>OVERVIEW</u>
4	INSTALLATION GUIDES
4	<u>INSTALLATION - GENERIC RESISTIVE</u>
5	<u>INSTALLATION - JVC</u>
6	<u>INSTALLATION - KENWOOD</u>
7	<u>INSTALLATION - ALPINE</u>
8	<u>INSTALLATION - PIONEER</u>
9	<u>INSTALLATION - USB HID</u>
10	<u>INSTALLATION - SONY</u>
11	<u>CONFIGURATION - HEADUNIT SELECTION</u>
12	<u>HEADUNIT LEGEND</u>
13	<u>DEFAULT FUNCTIONS</u>
14	<u>ASSEMBLY TIPS</u>
15	<u>VOLUME KNOB DIMENSIONS</u>
16	<u>TROUBLESHOOTING</u>
17	<u>RE_SWC SPECS</u>
18	<u>DISCLAIMERS</u>
19	<u>COMPLIANCE</u>
20	<u>DOCUMENT REVISION HISTORY</u>

OVERVIEW



1. UPDATE BUTTON

**2. WHITE STATUS
LIGHT**

**3. RED POWER
LIGHT**

**4. DIA 6mm
ENCODER SHAFT**

INSTALLATION - GENERIC RESISTIVE

- Solder the RED wire from the 2-pin connector on the RE_SWC kit to your headunit's steering wheel control input pin (generally labelled KEY1 or KEY2)
 - If you purchased the 3.5mm patch cable, this will be the tip of the TRS jack
- Solder the BLACK wire from the 2-pin connector into your harness if your headunit requires a reference for the SWC input (sometimes labelled ST REMOTE GND)
 - If you purchased the 3.5mm patch cable, this will be the sleeve of the TRS jack
- Power the kit with a USB C cable.
 - NOTE: This kit requires **5V USB power only**
- The RED power indicator on the RE_SWC will illuminate and stay on

Note: If your headunit requires holding the steering wheel control input when setting functions, double press the volume knob to enter learning mode (indicated with the WHITE status light flashing). The RE_SWC kit will hold the output for 4 seconds e.g. rotating the knob clockwise once will result in the RE_SWC kit holding the clockwise output for 4 seconds.

Repeat this process to program all functions

INSTALLATION - JVC

- Solder the RED wire from the 2-pin connector on the RE_SWC kit to your headunit's steering wheel control input pin (normally a blue and yellow wire)
 - If you purchased the 3.5mm patch cable, this will be the tip of the TRS jack
- Power the kit with a USB C cable.
 - NOTE: This kit requires **5V USB power only**
- The RED power indicator on the RE_SWC will illuminate and stay on
- The RE_SWC kit will operate without any further configuration on your headunit

INSTALLATION - KENWOOD

- Solder the RED wire from the 2-pin connector on the RE_SWC kit to your headunit's steering wheel control input pin (normally a blue and yellow wire)
 - If you purchased the 3.5mm patch cable, this will be the tip of the TRS jack
- Power the kit with a USB C cable.
 - NOTE: This kit requires **5V USB power only**
- The RED power indicator on the RE_SWC will illuminate and stay on
- The RE_SWC kit will operate without any further configuration on your headunit

INSTALLATION - ALPINE

- Plug the RE_SWC into your unit using the 3.5mm patch cable
 - You can make a compatible 3.5mm patch cable yourself by soldering the red wire from the 2-pin connector on the RE_SWC kit to the tip of the 3.5mm jack. Leave the ring and sleeve floating
- Power the kit with a USB C cable.
 - NOTE: This kit requires **5V USB power only**
- The RED power indicator on the RE_SWC will illuminate and stay on
- The RE_SWC kit will operate without any further configuration on your headunit

INSTALLATION - PIONEER

- Plug the RE_SWC into your unit using the 3.5mm patch cable
 - You can make a compatible 3.5mm patch cable yourself by soldering the red wire from the 2-pin connector on the RE_SWC kit to the tip of the 3.5mm jack. Leave the ring and sleeve floating
- Power the kit with a USB C cable.
 - NOTE: This kit requires **5V USB power only**
- The RED power indicator on the RE_SWC will illuminate and stay on
- Ensure “With Adapter” is set in the Steering Wheel Controls menu on your headunit
- The RE_SWC kit will operate without any further configuration on your headunit

INSTALLATION - USB HID

- Connect the RE_SWC kit to your device with a USB C cable (must support data & 5V power)
 - Note: Do not use this device with any power source higher than standard USB 5V
- The RED power indicator on the RE_SWC will illuminate and stay on

INSTALLATION - SONY

- Plug the RE_SWC into your unit using the 3.5mm patch cable
 - You can make a compatible 3.5mm patch cable yourself by soldering the red wire from the RE_SWC kit to the tip of the 3.5mm jack. Leave the ring and sleeve floating
- Power the kit with a USB C cable.
 - NOTE: This kit requires **5V USB power only**
- The RED power indicator on the RE_SWC will illuminate and stay on
- In the Steering Wheel Controls menu, set the [Custom] switch to [OFF]
 - On some units, you may need to set it to be “With Adapter”
- The RE_SWC kit will operate without any further configuration on your headunit

CONFIGURATION - HEADUNIT SELECTION

- Ensure RE_SWC is not powered
- Press and hold the volume knob. Connect power to the RE_SWC while keeping the volume knob held
- Release the volume knob after the WHITE status light is illuminated
- Press the volume knob once. The WHITE status light will flash once. Headunit number 1 is now programmed. Repeat this step until the desired headunit number is reached - refer to the headunit legend
- When the desired headunit number is entered, press and hold the volume knob. The WHITE status light will flash. Release the volume knob. The RE_SWC will flash x times where x is the headunit number entered

HEADUNIT LEGEND

HEADUNIT NUMBER	HEADUNIT TYPE / BRAND
1	GENERIC RESISTIVE
2	JVC
3	KENWOOD
4	ALPINE
5	PIONEER
6	USB HID
7	SONY

DEFAULT FUNCTIONS

<i>INPUT</i>	<i>OUTPUT</i>
<i>Volume Knob CW</i>	<i>Volume +</i>
<i>Volume Knob CCW</i>	<i>Volume -</i>
<i>Button Short Press</i>	<i>Mute</i>
<i>Button Long Press</i>	<i>Next Track</i>
<i>Button Double Press (Generic Resistive mode)</i>	<i>Enable Learning Mode</i>
<i>Button Double Press (Other Modes)</i>	<i>Previous Track</i>

ASSEMBLY TIPS

Note

- The RE_SWC will be shipped pre-configured in the mode selected during the checkout process for a plug-and-play setup

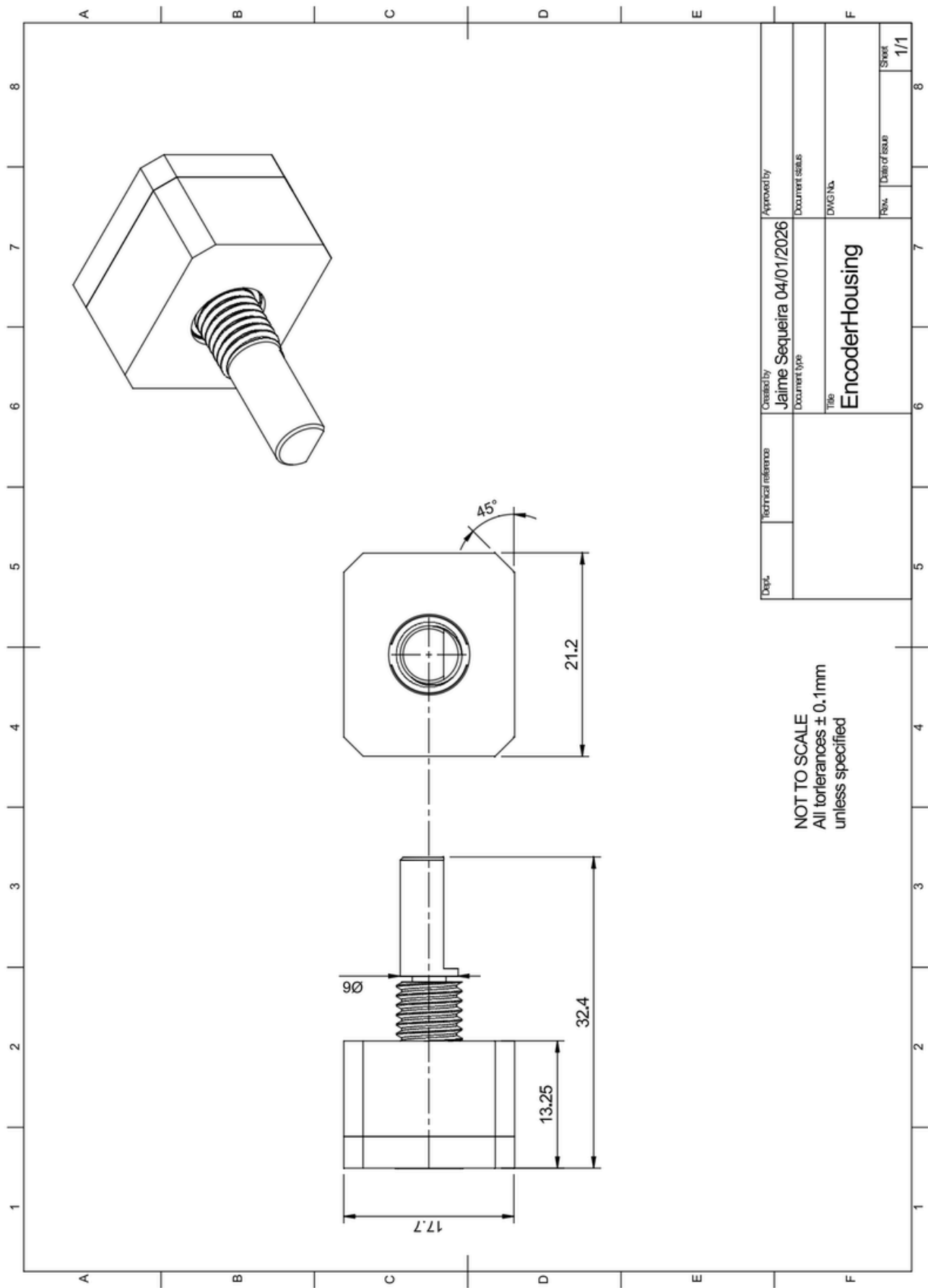
Mounting the knob

- Drill a pilot hole between 1 and 3mm in diameter
- Widen the hole to 7.5mm
- Mount the knob into the hole and secure with the included washer and nut

Powering the RE_SWC

- The RE_SWC uses 5V USB power. Do not use any higher voltage
- The RE_SWC boots up almost instantaneously. You can power the device with a switched 5V USB power supply. Configuration settings are retained between power cycles

VOLUME KNOB DIMENSIONS



TROUBLESHOOTING

- **The RED power light is off**
 - *Ensure you are powering the kit with 5V USB power*
- **The RED power light is on but the WHITE status light does not flash when I use the knob**
 - *Ensure the 5-pin connector harness for the knob is seated correctly and inserted fully into the knob and the kit*
- **The WHITE status light flashes when I turn the knob but my headunit does nothing**
 - *Ensure you have set the kit in the correct headunit mode*
 - *Ensure the RED wire from the 2-pin connector is soldered properly into your headunit's harness*
- **My headunit shows the volume increasing but the volume of my music doesn't change in USB HID mode**
 - *Some Android headunits do not route USB volume commands to the correct hardware. Revert to using the Resistive SWC mode with Key1 / Key 2 input wire instead*

RE_SWC SPECS

Input

Power: 5V / 100mA USB-C

SWC Header: **5V max.** Do not connect to 12V signal

CONNECTORS

VOLUME KNOB: JST SHR-05V-S-B

SWC OUT: JST SHR-02V-S-B

MCU:

WCH CH32X035F8U6

Digital Rheostat:

MCP4131-104E/SN

SPI @ 250kHz

DISCLAIMERS

- *Pounamu Electronics is not affiliated or associated with any headunit manufacturers or brands*
- *Pounamu Electronics is not affiliated or associated with the USB Implementers Forum (USB-IF)*

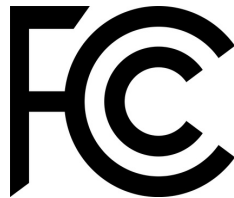
COMPLIANCE

FCC Compliance Statement

This device complies with Part 15 of the FCC Rules.

Operation is subject to the following two conditions:

- 1. This device may not cause harmful interference, and*
- 2. this device must accept any interference received, including interference that may cause undesired operation.*



Document Revision History

- *Rev 4*
 - *Added installation instructions for each headunit mode/brand*
- *Rev 3.2*
 - *Added separate Sony mode*
 - *Added warning not to exceed 5V on SWC input line*
- *Rev 3.1*
 - *Added troubleshooting content*
 - *Edited assembly instructions to include 5V power only spec*
- *Rev 3*
 - *Overhaul for RE_SWC V2*
 - *Added disclaimers section*
 - *Added compliance section*
- *Rev 2 - FW V2.0.0*
 - *Added instructions for JVC headunit interface*
 - *Added instructions for Kenwood headunit interface*
- *Rev 1 - Initial release*